



David Sarrat González

Bioinformatician

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About Me

I am a **Bioinformatician** and **Doctoral Researcher** specializing in federated analysis and standardized clinical data.

Research Interests:

- Federated data analysis
- OMOP CDM standardization
- **Privacy-preserving methodologies**
- Clinical research informatics

Current Focus: Developing innovative solutions for secure, collaborative analysis of sensitive healthcare data across multiple institutions.

Work Experience

	Doctoral Researcher Barcelona Institute for Global Health (ISGlobal)	Sep 2022 - Present
	Associate Professor Universitat de Lleida	Sep 2020 - Present
	Research Assistant Universitat de Lleida	Jun 2020 - Sep 2020
	Computer Engineering Intern Diputació de Lleida	Sep 2018 - Jun 2019
	Teacher Aula Técnica	Jun 2017 - Sep 2018

Education

	Doctor of Philosophy in Bioinformatics Universitat Autònoma de Barcelona	Sep 2022 - Present
	Master's degree in Data Science Universitat Oberta de Catalunya	Sep 2024 - Present
	Master's degree in Computer Engineering Universitat de Lleida	Sep 2019 - Jun 2022
	Bachelor's degree in Business Administration and Management Universitat Oberta de Catalunya	Sep 2016 - Jun 2021

Projects

dsOMOP: A DataSHIELD Integration for OMOP CDM Databases

A suite of DataSHIELD packages for integration with standardized clinical databases in the Observational Medical Outcomes Partnership Common Data Model (OMOP CDM).

Key features:

- Federated analysis capabilities
- **Secure data sharing** protocols
- Integration with OHDSI tools

Technologies used:

- R programming language
- **DataSHIELD framework**
- OMOP CDM standard

Technologies: R, DataSHIELD, OMOP CDM

Website: Project Documentation

Publications

dsOMOP: Bridging OMOP CDM and DataSHIELD for Secure Federated Analysis of Standardized Clinical Data

May 2025

David Sarrat-González, Xavier Escribà-Montagut, Jared Houghtaling, Juan R González

Bioinformatics

DOI: 10.1093/bioinformatics/btaf286

Predicting Injuries in Elite Female Football Players With Global-Positioning-System and Multiomics Data

May 2024

Juan R. González, Alejandro Cáceres, Eva Ferrer, Laura Balagué-Dobón, Xavier Escribà-Montagut, David Sarrat-González, Guillermo Quintás, Gil Rodas

International Journal of Sports Physiology and Performance

DOI: 10.1123/ijsp.2023-0184

Presentations

Integrating OMOP CDM and OHDSI Tools in Federated Clinical Data Analysis Frameworks

Jan 2024

Integraal Kankercentrum Nederland (IKNL) Seminar

Integrating DataSHIELD with OMOP CDM: A Novel Approach for Efficient Data Sharing and Analysis of Harmonized Clinical Data

Oct 2023

DataSHIELD Conference 2023

Workshops

Standardized Federated Data Analysis using DataSHIELD and dsOMOP

Jan 2025

Institut Català d'Oncologia (ICO)

Link: Workshop Materials

dsOMOP Software Demonstration: Hands-on Workshop for Federated Secure Analysis of Standardized Clinical Data in a COVID-19 Cohort

Jun 2024

OHDSI Europe Symposium 2024

Posters

dsRadiomics: A High-Performance Computing Framework for Federated Radiomic Feature Analysis Feb 2025

David Sarrat-González, Juan R González

Workshop of the Interuniversity PhD Programme in Bioinformatics 2025

dsOMOP: Federated Analysis of Harmonized Clinical Data Combining OMOP CDM and DataSHIELD in a DATOS-CAT Cohort Use Case Jun 2024

David Sarrat-González, Judith Martínez-González, Xavier Escribà-Montagut, Aikaterini Lymperidou, Ramón Mateo, Rafael de Cid, Juan R González, Alberto Labarga

OHDSI Europe Symposium 2024

DATOS-CAT: Methodologies for the standardization, integration and analysis of population-based biomedical data using semantic technologies Jun 2024

Judith Martinez Gonzalez, Guillem Bracons Cucó, Aikaterini Lymperidou, Xavier Escribà-Montagut, Marta Huertas, Ramon Mateo Navarro, David Sarrat González, Santiago Frid, Rafael de Cid, Juan Ramon González Ruiz, Alberto Labarga

15th International SWAT4HCLS Conference

DATOS-CAT: Leveraging ELIXIR ecosystem for the standardization, integration and federated analysis of population-based biomedical data May 2024

Judith Martinez-Gonzalez, David Sarrat-González, Ramon Mateo-Navarro, Guillem Bracons Cucó, Aikaterini Lymperidou, Marta Huertas, Sergi Aguiló-Castillo, Salvador Capella-Gutiérrez, Rafael de Cid, Santiago Frid, Juan R González, Alberto Labarga

ELIXIR All Hands 2024

RadGen4COPD: Developing a Secure and Efficient Platform for Integrated Medical Imaging and Genomic Studies of COPD Feb 2024

David Sarrat-González, Juan R González

Workshop of the Interuniversity PhD Programme in Bioinformatics 2024

Theses

DeliverEat: Design and Development of a Food Delivery Application Sep 2024

Tiberiu Ionut Paiu

Bachelor's Degree in Computer Engineering, Universitat de Lleida

Development and Integration of a Private Chatbot Using Retrieval-Augmented Generation Jun 2024

Laura Haro Escoi

Bachelor's Degree in Computer Engineering, Universitat de Lleida

URI: <https://repositori.udl.cat/handle/10459.1/466183>

Last Mile Delivery Optimization Jun 2024

Pol Escolà Curcó

Bachelor's Degree in Computer Engineering, Universitat de Lleida

URI: <https://repositori.udl.cat/handle/10459.1/466184>

Design and Development of a Farm Simulator Videogame Jun 2024

Maria Planella Vall

Bachelor's Degree in Computer Engineering, Universitat de Lleida

URI: <https://repositori.udl.cat/handle/10459.1/466020>

LifeTwin: A Unified Approach for Modelling Medical Digital Twins

Oct 2023

Josep Maria Salvia Hornos

Master's Degree in Computer Engineering, Universitat de Lleida

University Q&A Bot Using Contextualized GPT-3

Jun 2023

Marc Melis Batalla

Bachelor's Degree in Computer Engineering, Universitat de Lleida

URI: <https://repositori.udl.cat/handle/10459.1/463578>

Awards

HackEPS Winner of Group Saltó's Challenge

Nov 2022

HackEPS 2022

Development of a Real-Time Emotion Recognition System for an Assistant Robot using Deep Learning

Link: <https://devpost.com/software/emotion-recognizer-knj0ud>

Volunteering



President

IAESTE LC Lleida

Sep 2018 - Dec 2020



Co-Founder

LleidaHack

Apr 2018 - Dec 2020